Erdős problem 647: reductions, density bounds, and the prime-tuples barrier

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Abstract

Let τ be the divisor function. Erdős problem 647 asks whether there is an integer $n > 24$ with $\tau(n-k) \le k+2$ for all $1 \le k < n$; Erdős expected the answer to be no. Write $\mathcal{C}(x)$ for the number of such candidates $n \le x$. We give a reduction of the candidate set, the modular part of which is formally verified in Lean 4: every candidate $n > 84$ satisfies $2520 \mid n$ and, writing $n = 2520N$, the class of N modulo 46189 lies in an explicit set of 41; and every candidate $n > 24$ lies in one of two explicit prime-chain families. The families give, by an elementary Brun sieve, $|\mathcal{C}(x)| \ll x/(\log x)^7$; a separate growing-window argument gives $|\mathcal{C}(x)| \le x\exp(-(\log\log x)^{2-o(1)})$. Both are proved in the companion manuscripts. We then show why density methods stop here. The relaxation that drives them does not collapse, and its optimal output $x^{1-o(1)}$ cannot give finiteness; in the computed range, the genuine condition leaves only the known solution 24, and the deepest near-misses are prime chains. A uniform bound excluding arbitrarily long prime-chain near-misses would imply finiteness; such a bound appears to require input of prime-tuples type. The results are density-zero, not a resolution of the problem.

1 The problem

For a positive integer m let $\tau(m)$ be the number of its divisors. Erdős problem 647 [1] asks:

is there an integer $n > 24$ such that $\max\limits_{m<n}\big(m+\tau(m)\big) \le n+2$?

Equivalently, is there $n > 24$ with $\tau(n-k) \le k+2$ for every $1 \le k < n$. The bound $n+2$ is best possible, and $n = 24$ works; Erdős offered a prize for an example $n > 24$ and wrote that it is "extremely doubtful" that infinitely many exist, suggesting in fact that $\max_{m<n}(\tau(m)+m-n) \to \infty$. Call an integer $n > 24$ satisfying the condition a *candidate*, and let

$$\mathcal{C}(x) = \#\{\, n \le x : n \text{ is a candidate} \,\}.$$

This note collects, in one place, what the reduction of the candidate set buys. Two density bounds on $\mathcal{C}(x)$ are stated and attributed to the companion manuscripts [2, 3]; the modular reduction underlying them is formally verified [4]. The remaining sections then locate the obstruction to proving finiteness. None of this resolves the problem: every bound below is density-zero, and the step from density-zero to finiteness is traced to a prime-tuples-type obstruction.

Related work. The state of the art on prime factors of consecutive integers is Tao–Teräväinen [5], which resolves several neighbouring Erdős problems by the Maynard sieve and Pilatte's correlation estimates for bounded multiplicative functions, yet leaves #647 open, in the prime-tuples class. Adjacent results on divisor correlations in almost all short intervals [6] and on the distribution of $d(n+1)/d(n)$ [7] are average statements and do not bound the pointwise count $\mathcal{C}(x)$.

2 The reduction

Stage 1: modular constraints

The divisor conditions at the shifts k with small $\gcd(2520, k)$ force divisibility and exclude residue classes. The following is checked over the explicit finite search and is formally verified in Lean 4 against Mathlib; the verification artifact is the release `v0.1.0` of [4].

Theorem 2.1 (Stage-1 reduction; formally verified)**.** *Every candidate $n > 84$ satisfies $2520 \mid n$. Writing $n = 2520N$, the residue of N modulo $M = 11 \cdot 13 \cdot 17 \cdot 19 = 46189$ lies in an explicit set of 41 classes.*

The sieve modulo $11, 13, 17, 19$ first leaves 96 surviving classes; small-shift divisor-budget arguments close 55 of them, leaving the stated 41. The full deduction and its formalization are in [4]; the open 41-class set is listed there.

Stage 2: prime-chain families

Theorem 2.2 (Prime-chain reduction)**.** *Every candidate $n > 24$ lies in one of the two families*

$$n = 8s + 8, \quad s,\ 2s + 1,\ 4s + 3,\ 8s + 7 \text{ all prime},$$

or

$$n = 16s + 8, \quad s,\ 4s + 1,\ 8s + 3,\ 16s + 7 \text{ all prime}.$$

Sketch. The conditions at $k = 1, 2$ force $n = 2q + 2$ with q and $2q + 1$ prime. Writing $q - 1 = 2^a m$ with m odd, the condition at $k = 4$ forces $q = 2p + 1$ with $p, 2p + 1, 4p + 3$ prime. The condition at $k = 8$, with $p - 1 = 2^b s$, leaves exactly $b = 1$ and $b = 2$, the two displayed branches; the remaining cases are eliminated by the finite divisor-budget check carried out in [2]. □

3 Density bounds

Both families are admissible tuples of linear forms, so Brun's upper-bound sieve applies. Once $2520 \mid n$ (Theorem 2.1) the factor $3 \mid n$ is available, and the three further shifts $k \in \{3, 6, 12\}$ — whose 2520-forced part is exactly k — adjoin three more prime forms to each family. The seven prime-forcing shifts $\{1, 2, 3, 4, 6, 8, 12\}$ give, in each branch, an admissible seven-tuple (root-union $1, 2, 4, 6$ modulo $2, 3, 5, 7$, and at most $7 < p$ classes for $p > 7$).

Theorem 3.1 (Seven-form bound [2])**.** *$|\mathcal{C}(x)| \ll x/(\log x)^7$, unconditionally, with an effective absolute implied constant.*

The candidate condition can also be used only through the elementary implication $\tau(N) \geq \Omega(N) + 1$, where Ω counts prime factors with multiplicity. For a window of length H this gives $\Omega\big(\prod_{k=1}^{H}(n-k)\big) \leq H(H+3)/2$ for every candidate, and a Rankin moment together with the upper-bound β-sieve fundamental lemma yields a bound that improves on every fixed power of $\log x$. The companion growing-window manuscript proves the following.

Theorem 3.2 (Growing-window bound [3])**.** $|\mathcal{C}(x)| \leq x \exp\big(-(\log\log x)^{2-o(1)}\big)$.

Remark 3.3. Theorem 3.2 supersedes Theorem 3.1 and the earlier $x/(\log x)^k$ bounds: $\exp(-(\log\log x)^{2-o(1)})$ is smaller than $(\log x)^{-A}$ for every fixed A. We keep Theorem 3.1 because its proof is elementary and self-contained.

4 The finiteness barrier

Theorems 3.1 and 3.2 are density-zero results. We now explain why no refinement of this kind reaches finiteness, and where the genuine difficulty lies.

The density method cannot reach finiteness

The growing-window bound of Theorem 3.2 uses the candidate condition only through $\Omega(n-k) \le k+1$. For a window length H set

$$N_{\Omega,H}(x) = \#\{\, n \le x : \Omega(n-k) \le k+1 \text{ for } 1 \le k \le H \,\} \;\ge\; |\mathcal{C}(x)|.$$

Even a bound as strong as $x\exp(-c(\log\log x)^2)$, the natural frontier suggested by this relaxation, would still not imply finiteness.

Proposition 4.1. *$x\exp(-c(\log\log x)^2) = x^{1-o(1)} \to \infty$ for every $c > 0$. Hence no bound of this form, however sharp, implies $\mathcal{C}(x) = O(1)$.*

Proof. $(\log\log x)^2 = o(\log x)$, so $c(\log\log x)^2/\log x \to 0$ and the exponent of x tends to 1. □

Direct enumeration confirms that the relaxation itself does not collapse: $N_{\Omega,24}(10^7) = 208828$, and the count shows no sign of thinning to a finite set, whereas the genuine condition is far more rigid.

Enumeration of the genuine condition

Let $D(n)$ be the largest D with $\tau(n-k) \le k+2$ for all $1 \le k \le D$ (the depth of the longest run of satisfied conditions starting at n). A candidate is exactly an $n > 24$ with $D(n) \ge n-1$. Enumeration to 10^7 gives the following, computed independently with two divisor-function implementations.

D	$\#\{24 < n \le 10^7 : D(n) \ge D\}$
1	665014
2	30653
4	159
6	2
7	1
8	0

No $n > 24$ up to 10^7 survives eight consecutive conditions. The record depths grow far more slowly than n: the deepest runs below 10^7 reach

$$D(n) = 4,\ 5,\ 6,\ 7 \quad \text{at} \quad n = 720,\ 58680,\ 7224840,\ 9339120,$$

the integer attaining each new depth being far larger than the last. A candidate needs $D(n) \ge n-1$, whereas the deepest run anywhere below 10^7 has $D = 7$. The gap between the required depth and the observed record widens with n, the numerical form of Erdős's $\max_{m<n}(\tau(m) + m - n) \to \infty$.

What finiteness requires

The deepest near-misses are prime chains. At each record above, $n-1$ is prime, $n-2=2p$, $n-3=3p'$ and $n-4=2^2p''$ with p,p',p'' prime; deeper shifts admit other low-divisor shapes, and the run ends when a forced quotient first carries too many divisors. So large depth at these near-misses is governed by simultaneous prime-chain conditions of the type in Theorem 2.2.

This places the two directions of the problem on closely related prime-pattern objects. The existence of arbitrarily deep near-misses is naturally explained by prime-tuples heuristics, and Schinzel-type hypotheses predict such finite-window behaviour. Finiteness would require the opposite kind of input: a uniform obstruction preventing these prime-chain patterns from extending far enough. Such an obstruction is not supplied by the density estimates above, and appears to lie in the same prime-tuples barrier discussed for related divisor problems by Tao and Teräväinen [5]. The reduction here makes that barrier visible; it does not cross it.

5 Formal verification and reproducibility

Theorem 2.1 is formalized in Lean 4 against Mathlib in the public repository [4], release `v0.1.0`. The build is free of `sorry`; an axiom audit accompanies the release, recording the single project-level axiom and the standard foundations on which the verified statements depend. The enumeration of Section 4 and the admissibility checks of Section 3 are reproduced by the scripts in the same repository.

Acknowledgments

The partial modular observations behind Theorem 2.1 are due to S. Dutta and B. Alexeev in the erdosproblems.com discussion of the problem [1], and the clean shift $k=3$ underlying the seven-form bound (Theorem 3.1) is due to K. Kitamura. The formal verification uses Lean 4 and Mathlib; numerical and symbolic checks used PARI/GP, Wolfram, and SymPy. AI-based tools assisted with drafting and exploratory checking, and all mathematical content was verified independently.

References

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